

## Practical - 5

**Aim:** Apply different features of Express.js to make web applications using Node.js

**1:** Build a simple express js program to add, update, delete and display all products available in list of e-commerce application using get, put, post and delete methods.

Assume that products are available on products object. Product properties are product\_id, product\_name, product\_size, product\_brand, product\_color.

**Code:**

```
const express = require('express');

const app = express();
app.use(express.json());

// Products object/list
let products = [
  {
    product_id: 1,
    product_name: 'T-Shirt',
    product_size: 'L',
    product_brand: 'Nike',
    product_color: 'Black',
  },
  {
    product_id: 2,
    product_name: 'Jeans',
    product_size: 'M',
    product_brand: 'Levis',
    product_color: 'Blue',
  },
];

app.get('/products', (req, res) => {
  res.json(products);
});

app.post('/products', (req, res) => {
  const newProduct = {
    product_id: products.length + 1,
    product_name: req.body.product_name,
    product_size: req.body.product_size,
    product_brand: req.body.product_brand,
    product_color: req.body.product_color,
  };

  products.push(newProduct);
```

```
res.status(201).json({
  message: 'Product added successfully',
  product: newProduct,
});
});

app.put('/products/:id', (req, res) => {
  const id = parseInt(req.params.id);

  const product = products.find((p) => p.product_id === id);

  if (!product) {
    return res.status(404).json({
      message: 'Product not found',
    });
  }

  product.product_name = req.body.product_name;
  product.product_size = req.body.product_size;
  product.product_brand = req.body.product_brand;
  product.product_color = req.body.product_color;

  res.json({
    message: 'Product updated successfully',
    product: product,
  });
});

app.delete('/products/:id', (req, res) => {
  const id = parseInt(req.params.id);

  const index = products.findIndex((p) => p.product_id === id);

  if (index === -1) {
    return res.status(404).json({
      message: 'Product not found',
    });
  }

  const deletedProduct = products.splice(index, 1);
  res.json({
    message: 'Product deleted successfully',
    product: deletedProduct[0],
  });
});

app.listen(3000, () => {
  console.log('Server running at http://localhost:3000');
});
```

**Output:**

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-5>node Practical_5_1.js
Server running at http://localhost:3000
```

**GET** http://localhost:3000/products:

**Response:**

```
[
  {
    "product_id": 1,
    "product_name": "T-Shirt",
    "product_size": "L",
    "product_brand": "Nike",
    "product_color": "Black"
  },
  {
    "product_id": 2,
    "product_name": "Jeans",
    "product_size": "M",
    "product_brand": "Levis",
    "product_color": "Blue"
  }
]
```

**POST** http://localhost:3000/products:

**Body:**

```
{
  "product_name": "Shirt",
  "product_size": "L",
  "product_brand": "Polo",
  "product_color": "Maroon"
}
```

**Response:**

```
{
  "message": "Product added successfully",
  "product": {
    "product_id": 3,
    "product_name": "Shirt",
    "product_brand": "Polo",
    "product_color": "Maroon"
  }
}
```

**POST** http://localhost:3000/products/1:

**Body:**

```
{
  "product_name": "T-Shirt",
  "product_size": "L",
  "product_brand": "Nike",
  "product_color": "White"
}
```

**Response:**

```
{
  "message": "Product updated successfully",
  "product": {
```

```

        "product_id": 1,
        "product_name": "T-Shirt",
        "product_size": "L",
        "product_brand": "Nike",
        "product_color": "White"
    }
}

```

**DELETE** http://localhost:3000/products/1:

**Response:**

```

{
  "message": "Product deleted successfully",
  "product": {
    "product_id": 1,
    "product_name": "T-Shirt",
    "product_size": "L",
    "product_brand": "Nike",
    "product_color": "White"
  }
}

```

**2:** Create express js program using router for e-commerce application. This application has 4 modules: order, user, product and category. Each module has 4 methods, namely get, put, post and delete and has path '/', '/update-details', 'create-details', and 'delete-details respectively. For example: 'localhost:8000/product' will be accessed and the page '/' will displayed message "get method from user module", same for other 3 methods and modules.

**Code:**

```

const express = require('express');

const app = express();

app.use(express.json());

// Import routers
const orderRouter = require('./routes/order');
const userRouter = require('./routes/user');
const productRouter = require('./routes/product');
const categoryRouter = require('./routes/category');

// Mount routers
app.use('/order', orderRouter);
app.use('/user', userRouter);
app.use('/product', productRouter);
app.use('/category', categoryRouter);

// Start server
app.listen(8000, () => {
  console.log('Server running at http://localhost:8000');
});

```

**Code category.js:**

```
const express = require('express');

const router = express.Router();
router.get('/', (req, res) => {
  res.send('get method from category module');
});
router.put('/update-details', (req, res) => {
  res.send('put method from category module');
});
router.post('/create-details', (req, res) => {
  res.send('post method from category module');
});
router.delete('/delete-details', (req, res) => {
  res.send('delete method from category module');
});
module.exports = router;
```

**Code order.js:**

```
const express = require('express');

const router = express.Router();
router.get('/', (req, res) => {
  res.send('get method from order module');
});
router.put('/update-details', (req, res) => {
  res.send('put method from order module');
});
router.post('/create-details', (req, res) => {
  res.send('post method from order module');
});
router.delete('/delete-details', (req, res) => {
  res.send('delete method from order module');
});
module.exports = router;
```

**Code product.js:**

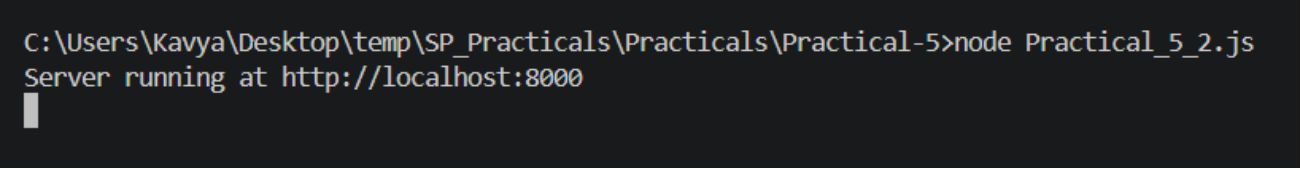
```
const express = require('express');

const router = express.Router();
router.get('/', (req, res) => {
  res.send('get method from product module');
});
router.put('/update-details', (req, res) => {
  res.send('put method from product module');
});
router.post('/create-details', (req, res) => {
  res.send('post method from product module');
```

```
});  
router.delete('/delete-details', (req, res) => {  
  res.send('delete method from product module');  
});  
module.exports = router;
```

**Code user.js:**

```
const express = require('express');  
  
const router = express.Router();  
router.get('/', (req, res) => {  
  res.send('get method from user module');  
});  
router.put('/update-details', (req, res) => {  
  res.send('put method from user module');  
});  
router.post('/create-details', (req, res) => {  
  res.send('post method from user module');  
});  
router.delete('/delete-details', (req, res) => {  
  res.send('delete method from user module');  
});  
module.exports = router;
```

**Output:**

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-5>node Practical_5_2.js  
Server running at http://localhost:8000
```

**GET** http://localhost:8000/category:

**Response:**

get method from category module

**POST** http://localhost:8000/category/create-details:

**Response:**

post method from category module

**PUT** http://localhost:8000/category/update-details:

**Response:**

put method from category module

**DELETE** http://localhost:8000/category/delete-details:

**Response:**

delete method from category module

**GET** http://localhost:8000/order:

**Response:**

get method from order module

**POST** http://localhost:8000/order/create-details:

**Response:**

post method from order module

**PUT** http://localhost:8000/order/update-details:

**Response:**

put method from order module

**DELETE** http://localhost:8000/order/delete-details:

**Response:**

delete method from order module

**GET** http://localhost:8000/product:

**Response:**

get method from product module

**POST** http://localhost:8000/product/create-details:

**Response:**

post method from product module

**PUT** http://localhost:8000/product/update-details:

**Response:**

put method from product module

**DELETE** http://localhost:8000/product/delete-details:

**Response:**

delete method from product module

**GET** http://localhost:8000/user:

**Response:**

get method from user module

**POST** http://localhost:8000/user/create-details:

**Response:**

post method from user module

**PUT** http://localhost:8000/user/update-details:

**Response:**

put method from user module

**DELETE** http://localhost:8000/user/delete-details:

**Response:**

delete method from user module

**3:** Create express js program that employ middleware using next() function.

**Code:**

```
const express = require('express');

const app = express();

// Middleware 1
function middleware1(req, res, next) {
  console.log('Middleware 1 executed');
  next();
}

// Middleware 2
function middleware2(req, res, next) {
  console.log('Middleware 2 executed');
  next();
}

// Register middleware
app.use(middleware1);
app.use(middleware2);

// Route
app.get('/', (req, res) => {
  res.send('Welcome to Express.js');
});

// Start server
app.listen(8000, () => {
  console.log('Server running at http://localhost:8000');
});
```

**Output:**

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-5>node Practical_5_3.js
Server running at http://localhost:8000
```

**GET** http://localhost:8000:

**Response:**

Welcome to Express.js

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-5>node Practical_5_3.js
Server running at http://localhost:8000
Middleware 1 executed
Middleware 2 executed
```